1) If step size is very large in RK4, result will be:

A) More accurate

B) Less accurate

C) Exact

D) Infinite

2) . In RK4 method, k_2 and k_3 are $calculated$ at;

A) Initial point

B) Final point

C) Midpoint

D) Random point

3) In trapezoidal rule, increasing number of subintervals leads to:

A) Decrease in accuracy

B) Increase in accuracy

C) No change

D) Infinite error

4) In Newton-Raphson method, if f''(x) is large near the root, the method:

A) Converges very fast

B) May become unstable

C) Becomes exact

D) No effect

5) Higher order derivative increases:

A) Accuracy

B) Error

C) Stability

D) None

6) In numerical differentiation, small error in data leads to:

A) Small output error

B) Large output error

C) No effect

D) Exact result

7) If matrix is not diagonally dominant, Gauss-Seidel method:

A) Always converges

B) May diverge

C) Always exact

D) No solution

8) Jacobi method differs from Gauss-Seidel because it uses:

A) Updated values immediately

B) Only previous iteration values

C) Both

D) None

9) Modified Euler method improves accuracy by:

A) Using midpoint

B) Using derivative twice

C) Using integration

D) None

10) RK4 method is based on:

A) Taylor series approximation

B) Fourier series

C) Laplace transform

D) Matrix theory

11) In Newton-Raphson method, if initial guess is very far from root, method may:

A) Converge faster

B) Diverge

C) Become exact

D) No effect

12) Rate of convergence of Secant method is approximately:

A) 1

B) 1.618

C) 2

D) 3

13) Secant method requires:

A) One initial guess

B) Two initial guesses

C) Three initial guesses

D) None

14) If f(x) is discontinuous, bisection method:

A) Always works

B) May fail

C) Always converges

D) Gives exact solution

15) In interpolation, oscillation error is called:

A) Truncation error

B) Runge phenomenon

C) Round-off error

D) Absolute error

16) Degree of polynomial passing through 5 points:

A) 3

B) 4

C) 5

D) 6

17) Central difference formula is derived from:

A) Forward difference

B) Backward difference

C) Both

D) None

18) If $\Delta f(x)$=constant, function is:

A) Linear

B) Quadratic

C) Cubic

D) Exponential

19) Composite trapezoidal rule improves accuracy by:
A) Increasing h
B) Decreasing h
C) Keeping h constant
D) None

20) If function is highly oscillatory, numerical integration becomes:
A) Easier
B) Difficult
C) Exact
D) Zero

21) If step size h is reduced, RK4 accuracy:
A) Decreases
B) Increases
C) Same
D) Zero

22) RK4 method global error order is:
A) $O(h^2)$
B) $O(h^3)$
C) $O(h^4)$
D) $O(h^5)$

23) Which combination is correct in RK4 formula?
A) $k_1 + k_2 + k_3 + k_4$
B) $k_1 + 2k_2 + 2k_3 + k_4$
C) $2k_1 + k_2 + k_3 + k_4$
D) $k_1 + k_4$

24) Given $\frac{dy}{dx} = x^2 + y \, , y(0) = 1, h = 0.1, find\ k_1$
A) 0.1
B) 0.01
C) 0.11
D) None of the above

25) Given $\frac{dy}{dx} = x \ + y \, , y(0) = 1, h = 0.1, find\ k_1$
A) O.1
B) 0.2
C) 0.05
D) 1

26) Bisection method requires:
A) f(a) = f(b)
B) f(a)·f(b) < 0
C) f(a)·f(b) > 0
D) None

27) Order of convergence of Bisection method:
A) 2
B) 1
C) 3
D) 0

28) Newton-Raphson method is based on:

A) Taylor expansion

B) Fourier series

C) Laplace transform

D) Euler method

29) Newton method formula:

A) $X_n + 1 = X_n + \frac{f(X_n)}{f`(x_n)}$

B) $X_n + 1 = X_n - \frac{f(X_n)}{f`(x_n)}$

C) $X_n + 1 = \frac{f(X_n)}{f`(x_n)}$

D) None

30) Order of convergence of Newton's method:

A) 1

B) 2

C) 3

D) 4

31) False position method is:

A) Open method

B) Bracketing method

C) Iterative method only

D) None

32) If f'(x)=0 near root, Newton method:

A) Converges fast

B) Fails

C) Becomes linear

D) Oscillates

33) Rate of convergence depends on:

A) Initial guess

B) Function smoothness

C) Multiplicity of root

D) All

34) Bisection error after n iterations:

A) $\frac{(b-a)}{2^n}$

B) $\frac{(b-a)}{2}$

C) n(b−a)

D) None

35) Multiple root slows:

A) Bisection

B) Newton method

C) False position

D) All

36) Forward difference operator:

A) ∇

B) Δ

C) E

D) D

37) Backward difference operator:
   A) Δ
   B) ∇
   C) E
   D) D
38) Shift operator E:
   A) f(x+h)
   B) f(x−h)
   C) f(x+h)−f(x)
   D) None
39) Δf(x) equals:
   A) f(x+h)−f(x)
   B) f(x)−f(x−h)
   C) f(x+h)+f(x)
   D) None
40) Newton forward formula used when:
   A) x near end
   B) x near beginning
   C) x anywhere
   D) None
41) Newton backward used when:
   A) Start
   B) Middle
   C) End
   D) None
42) Divided difference used for:
   A) Equal interval
   B) Unequal interval
   C) Both
   D) None
43) Lagrange interpolation is:
   A) Numerical integration
   B) Polynomial interpolation
   C) Root finding
   D) None
44) Degree of interpolating polynomial for n+1 points:
   A) n
   B) n+1
   C) n−1
   D) 2n
45) Gauss formula is:
   A) Forward
   B) Central
   C) Backward
   D) None
46) Stirling formula used for:
   A) Beginning
   B) End

C) Middle

D) None

47) Bessel formula is used when:

A) Even number of data

B) Odd

C) Unequal

D) None

48) Error in interpolation depends on:

A) Step size

B) Degree

C) Smoothness

D) All

49) Finite differences vanish for polynomial degree n at:

A) n

B) n+1

C) n−1

D) 2n

50) Laplace-Everett formula is:

A) Central

B) Forward

C) Backward

D) None

51) Forward difference is less accurate than central difference:

A) True

B) False

C) Same accuracy

D) Cannot say

52) Backward difference is mainly used near the end of data:

A) True

B) False

C) Used at beginning

D) Not used

53) Divided difference table is:

A) Triangular

B) Square

C) Rectangular

D) Diagonal

54) . Lagrange interpolation formula is:

A) Symmetric

B) Asymmetric

C) Linear

D) Discontinous

55) Error term in interpolation involves:

A) Higher order derivative

B) Lower derivative

C) No derivative

D) Constant

56) . $\Delta^2 f(x)$ represents:
A) First difference
B) Second difference
C) Third difference
D) No difference
57) Interpolation is exact if function is:
A) Polynomial of degree $\leq n$
B) Exponential
C) Trigonometric
D) Logarithmic
58) Central difference formulas generally:
A) Increase error
B) Reduce error
C) No change
D) Make unstable
59) Unequal interval interpolation uses:
A) Newton forward
B) Newton backward
C) Newton divided difference
D) Gauss formula
60) Step size (h) affects:
A) Accuracy
B) Speed
C) Both
D) None
61) Numerical differentiation is based on:
A) Differences
B) Integration
C) Limits
D) Series
62) Most accurate difference formula:
A) Forward
B) Backward
C) Central
D) None
63) Trapezoidal rule approximates using:
A) Straight line
B) Parabola
C) Cubic
D) Circle
64) Simpson 1/3 rule requires:
A) Even intervals
B) Odd
C) Any
D) Prime
65) Error in trapezoidal rule depends on:
A) $f'(x)$
B) $f''(x)$

C) $f'''(x)$

D) $f^{(4)}(x)$

66) Simpson’s rule is more accurate because it uses:

A) Linear approximation

B) Quadratic approximation

C) Cubic

D) Constant

67) Maximum value occurs when:

A) $f'=0$ and $f''>0$

B) $f'=0$ and $f''<0$

C) $f'\neq 0$

D) $f''=0$

68) Minimum value occurs when:

A) $f'=0$ and $f''>0$

B) $f'=0$ and $f''<0$

C) $f'\neq 0$

D) $f''=0$

69) Numerical integration gives:

A) Exact value

B) Approximate value

C) Infinite value

D) Zero

70) Simpson’s 1/3 rule is exact for:

A) Linear function

B) Quadratic

C) Cubic polynomial

D) Quartic

71) Trapezoidal rule is exact for:

A) Linear function

B) Quadratic

C) Cubic

D) Quartic

72) Numerical differentiation is:

A) Stable

B) Sensitive to errors

C) Exact

D) Simple always

73) Numerical integration is:

A) Sensitive to error

B) Smoothens error

C) Unstable

D) Divergent

74) Central difference formula error is:

A) $O(h)$

B) $O(h^2)$

C) $O(h^3)$

D) $O(h^4)$

75) Trapezoidal rule error is:

A) O(h)

B) $O(h^2)$

C) $O(h^3)$

D) $O(h^4)$

76) Simpson's 1/3 rule error is:

A) $O(h^2)$

B) $O(h^3)$

C) $O(h^4)$

D) $O(h^5)$

77) Step size decreases → accuracy:

A) Decreases

B) Increases

C) Same

D) Zero

78) . Integration approximates:

A) Area

B) Volume

C) Length

D) None

79) Differentiation approximates:

A) Area

B) Slope

C) Volume

D) None

80) Simpson's rule requires:

A) Even intervals

B) Odd

C) Any

D) Prime

81) Numerical methods are used when:

A) Exact solution difficult

B) Exact solution easy

C) No equation

D) Always exact

82) Forward difference accuracy is:

A) High

B) Low

C) Same

D) Infinite

83) Backward difference is similar to:

A) Forward difference

B) Central difference

C) Simpson

D) None

84) Increasing h leads to:

A) Better accuracy

B) Poor accuracy

C) No effect
D) Infinite accuracy

85) Numerical integration methods are based on:
A) Geometry
B) Algebra
C) Calculus
D) None

86) Simpson rule uses:
A) Straight line
B) Parabola
C) Circle

87) Trapezoidal rule divides area into:
A) Rectangles
B) Triangles
C) Trapeziums
D) Circles

88) Trapezoidal rule divides area into:
A) Rectangles
B) Triangles
C) Trapeziums
D) Circles

89) Polynomial approximation helps in:
A) Integration
B) Differentiation
C) Interpolation
D) All

90) Euler method approximates:
A) Algebraic eqn
B) ODE
C) Matrix
D) Integral

91) Euler method is:
A) First order
B) Second order
C) Third order
D) Fourth order

92) Picard method is based on:
A) Differentiation
B) Integration
C) Matrix
D) Series

93) RK4 method order is:
A) 1
B) 2
C) 3
D) 4

94) RK4 method is:
A) Less accurate

B) Highly accurate
C) Same as Euler
D) Divergent

95) Gauss elimination converts system into:
A) Diagonal form
B) Upper triangular form
C) Lower triangular
D) Identity

96) Back substitution is used after:
A) Iteration
B) Elimination
C) Integration
D) Differentiation

97) Pivoting is used to:
A) Increase error
B) Reduce error
C) No effect
D) Remove equation

98) Gauss-Seidel method is faster than:
A) Jacobi method
B) Euler method
C) RK method
D) None

99) Convergence depends on:
A) Initial guess
B) Matrix
C) Both
D) None

100) Diagonal dominance ensures:
A) Divergence
B) Convergence
C) Oscillation
D) None

101) Picard method gives:
A) Exact solution
B) Series solution
C) Numerical only
D) Matrix

102) Euler method has:
A) High error
B) Low error
C) No error
D) Infinite accuracy

103) Step size decrease → Euler accuracy:
A) Decreases
B) Increases
C) Same
D) Zero

104) RK4 uses:

A) One slope

B) Two slopes

C) Four slopes

D) Infinite

105) Linear system written as:

A) AX = B

B) A+B

C) X+A

D) None

106) Unique solution exists if:

A) det(A)=0

B) det(A)≠0

C) det(A)=1

D) None

107) Iterative methods may:

A) Always converge

B) Diverge

C) Exact always

D) None

108) Stability in numerical methods means:

A) Error control

B) No error

C) Infinite error

D) None

109) Round-off error occurs due to:

A) Approximation

B) Machine precision

C) Both

D) None

110) Truncation error occurs due to:

A) Finite steps

B) Infinite steps

C) Exact solution

D) None

111) RK4 is preferred because:

A) Simple

B) Accurate

C) Fast

D) None

112) Condition number measures:

A) Stability

B) Error sensitivity

C) Both

D) None

113) Picards method use to solve:

A. BVP

B. IVP

C. Algebraic equation

D. PDE

114) Simpson 3/8 rule requires:
A) Multiple of 3
B) Even
C) Odd
D) Any

115) Gauss elimination is:
A) Direct method
B) Iterative
C) Both
D) None

116) Gauss-Seidel method is:
A) Iterative
B) Direct
C) Analytical
D) None

117) convergence condition:
A) Diagonal dominance
B) Symmetry
C) Both
D) None

118) Euler method is used for:
A) Algebraic eqn
B) ODE
C) PDE
D) Matrix

119) Euler formula:
A) $y_{n+1} = y_n + hf(x,y)$
B) $y_{n+1} = y_n - hf(x,y)$
C) $y_{n+1} = hf$
D) None

120) Using bisection method, root of $x^3 - x - 2 = in$ [1,2] after 1 iteration is:
A) 1.25
B) 1.5
C) 1.75
D) 2

121) Which of the following is the correct formula for Picard's method?

A) $y_{n+1} = y_n + hf(x_n, y_n)$
B) $y_{n+1} = y_0 + \int_{x_0}^{x} f\big(t, y_n(t)\big)dt$
C) $y_{n+1} = y_n + \frac{h}{2}(k_1 + k_2)$
D) $y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_3)$

122) The interval of integration in Picard method is:

A) 0 to 1
B) x to y
C) $x_0 \; to \; x$
D) infinite

123) If $f(x)=x^2$, $\Delta f(1)$ =

A) 2
B) 3
C) 4
D) 5

124) Finite differences of degree n vanish at:

A) n
B) n+1
C) n-1
D) 2n

125) The first approximation $y_1(x)$ depends on:

A) y_1
B) y_2
C) y_0
D) $none$